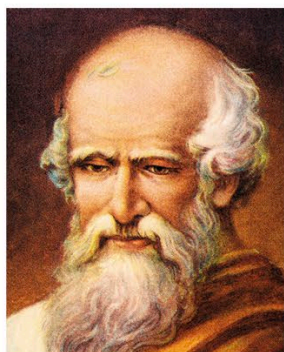


Archimedes



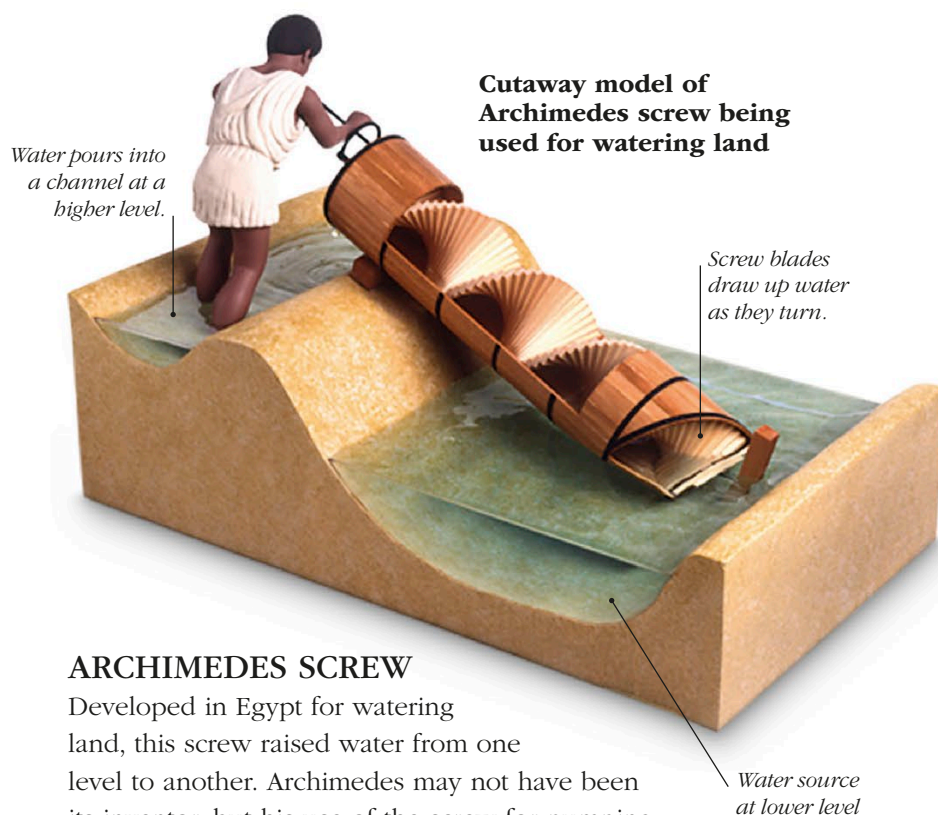
One of the greatest ancient inventors, Archimedes was also a brilliant mathematician and physicist. Born in the Greek city of Syracuse in Sicily in 287 BCE, it is believed he was sent to Egypt to be educated. Archimedes invented many important mechanical devices, including a pulley system that could lift enormous weights. His scientific writings are still useful to scholars today.

FIGHTING THE ROMANS

When Rome attacked Syracuse in 214 BCE, it is believed that two of Archimedes' inventions were used to defend his home city. The first used mirrors to focus the sun's rays on the Roman ships and set fire to them. The second, called the "iron hand" (below), was a crane with a giant grappling hook that could grab the ships and capsize them.



Italian painting showing Archimedes' grappling hook as a huge hand, 1600



ARCHIMEDES SCREW

Developed in Egypt for watering land, this screw raised water from one level to another. Archimedes may not have been its inventor, but his use of the screw for pumping water out of ships made the device well known. The screw was like a corkscrew inside a hollow cylinder. As it turned, it drew water upward.

LIFE STORY

287 BCE

Archimedes is born in Syracuse, Sicily, at the time a Greek city-state. His father was an astronomer and mathematician.

c. 250 BCE

He is believed to go to Egypt to study. Archimedes writes a paper on geometry called *On the Measurement of a Circle*, and another about fluids called *On Floating Bodies*.

c. 225 BCE

He writes two other important works: *On Spirals* and *On the Sphere and the Cylinder*.

218 BCE

Second Punic War begins between Carthage and Rome. Archimedes' home, Syracuse, allies with Carthage.